



Mbemba Ngum

Climate Data Specialist | Meteorologist | Agrometeorologist

Department of Water Resources

Banjul, The Gambia

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🇬🇲 Gambian

Research Interests

- 📊 Climate Data Management

Professional Summary

Climate Data Specialist, Meteorologist and Agrometeorologist with over six years of professional experience in climate data management, meteorological analysis, seasonal forecasting, and climate information services. Experienced in collecting, quality controlling, validating, harmonizing and analysing meteorological and climatological datasets to support climate monitoring, agricultural planning, disaster risk reduction and evidence-based decision making.

Skilled in Python, R, GIS, remote sensing, scientific programming, climate statistics and environmental data visualization. Experienced in developing climate information products, seasonal forecasts, agrometeorological bulletins, climate maps and technical reports for national and international stakeholders.

Education

- **Master of Engineering in Agrometeorology**

AGRHYMET Regional Climate Center for West Africa and the Sahel, Niamey, Niger
2020 – 2024

Thesis: Predictors Adapted to Forecasting Agroclimatic Characteristics for The Gambia

-  Climate Information Services
-  Climate Data Analytics
-  Seasonal Climate Forecasting
-  Earth Observation & Remote Sensing
-  GIS & Spatial Analysis
-  Climate Risk Assessment
-  Climate Change Adaptation
-  Climate-Smart Agriculture
-  Artificial Intelligence for Climate Services

Languages

-  Mandinka (Native)
-  Wolof (Native)
-  English (C2)
-  French (C2 Speaking/Reading, B2 Writing)
-  Arabic (Basic)

Technical Tools

-  Python (Pandas, NumPy, Xarray, GeoPandas)
-  R Programming
-  VS Code
-  JupyterLab
-  Git & GitHub
-  ArcGIS
-  Google Earth Engine
-  SEPAL
-  CDO & NCO
-  WASS2S Forecasting System

- **Medium-Level Technician in Meteorology (WMO Class III)**
Nigerian Meteorological Agency (NiMet), Nigeria
2018 – 2019
Thesis: Impact of Climate Change on Fishing Activities in The Gambia
- **BSc Economics & Management**
University of The Gambia
2015 – 2018

Professional Experience

- **Meteorologist / Research Analyst**
Department of Water Resources, Banjul
2019 – Present
 - Manage and analyse national meteorological and climatological datasets.
 - Perform climate data quality control, validation and interpretation.
 - Conduct operational weather forecasting and seasonal climate analysis.
 - Contribute to seasonal forecasting of rainfall quantity and agroclimatic parameters.
 - Develop agrometeorological bulletins, climate reports and advisory products.

- 🌤️ CPT & PyCPT
- 🗺️ GrADS & Panoply
- ☁️ ERA5
- 🌧️ TAMSAT
- 🌾 AquaCrop

- Produce GIS-based climate maps and spatial analyses.
- Develop scientific workflows using Python, R, VS Code and JupyterLab.
- Support climate services for agriculture, water resources management and early warning systems.

- **Geography & Agricultural Science Teacher**

Masroor & Da Vinci Senior Secondary Schools
2013 – 2015

Selected Technical Projects

- Development of a Quality-Controlled National Rainfall Dataset for The Gambia (1981–2020).
- Quality Control, Homogeneity Assessment and Reconstruction of Historical Rainfall Records.
- Seasonal Forecasting of Agroclimatic Predictors using the WASS2S Forecasting System.
- Climate Data Analysis and Visualization using Python and GIS.
- Spatial and Temporal Analysis of Rainfall Variability across The Gambia.
- Drought Monitoring using Standardized

Precipitation Index (SPI).

Technical Skills

- **Climate Data Management:** Quality Control, Validation, Homogeneity Testing, Missing Data Reconstruction
- **Programming & Data Science:** Python, R, Git, VS Code, JupyterLab
- **Climate Analysis:** Time Series Analysis, Climate Statistics, Forecast Verification
- **GIS & Remote Sensing:** ArcGIS, Google Earth Engine, SEPAL, GeoPandas
- **Climate Datasets:** ERA5, TAMSAT, CHIRPS, CORDEX-Africa, Station Observations
- **Climate Visualization:** Matplotlib, Panoply, Climate Mapping

Trainings & Professional Development

- Air Quality and Climate Training – Mohammed VI Polytechnic University (UM6P), Morocco (2025)
- Hydrological Forecasting Training – UKCEH/WASCAL, Accra, Ghana (2026)
- AGRHYMET Seasonal & Sub-seasonal Forecasting Training (2024)
- Python for Meteorological Applications (2024)
- Climate Downscaling & Decision Support –

GCCA+ (2024)

- GSARS MSF Training – FAO (2024)

Professional Roles

- National Focal Point – New Generation Seasonal & Sub-seasonal Forecasting (AGRHYMET RCC-WAS)
- National Youth Focal Point – WMO Youth Network

Professional Memberships

- World Meteorological Organization (WMO) Youth Network
- AGRHYMET Regional Climate Centre Community
- Department of Water Resources Meteorological Unit

References

- **Dr. Tinni Halidou Seydou**
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